

Learning dynamics of Deep Linear Networks

Exercise Session — 1h30, pen and paper

Use LLMs to help you out when you spend longer than the suggested time (but make sure that you understand). Make sure you keep at least 30 minutes to do problem 3. **References:**

- Saxe, McClelland & Ganguli, *Exact solutions to the nonlinear dynamics of learning in deep linear neural networks*, ICLR 2014 / PNAS 2019
- Achour, Malgouyres & Gerchinovitz, *The loss landscape of deep linear neural networks: a second-order analysis*, JMLR 2024
- Tu, Aranguri & Jacot, *Mixed Dynamics In Linear Networks: Unifying the Lazy and Active Regimes*, 2024

Setup and notation

We study deep linear networks (DLNs) with L weight matrices $W_1 \in \mathbb{R}^{d_1 \times d_0}$, $W_2 \in \mathbb{R}^{d_2 \times d_1}$, $\dots$, $W_L \in \mathbb{R}^{d_L \times d_{L-1}}$. The network computes:

$$f(x) = W_L W_{L-1} \cdots W_1 x =: Wx$$

where $W = W_L \cdots W_1 \in \mathbb{R}^{d_L \times d_0}$ is the end-to-end (or “student”) matrix. We train on a dataset $\{(x_\mu, y_\mu)\}_{\mu=1}^N$ with the squared loss:

$$\mathcal{L}(\theta) = \frac{1}{2N} \sum_{\mu=1}^N \|y_\mu - W_L \cdots W_1 x_\mu\|^2$$

In the population limit with whitened inputs $\Sigma_X = I$, this becomes:

$$\mathcal{L}(W) = \frac{1}{2} \|M - W\|_F^2$$

where $M = \Sigma_{YX} \Sigma_X^{-1}$ is the “teacher” matrix (the OLS solution) with SVD $M = U \text{diag}(s_1, \dots, s_r, 0, \dots, 0) V^\top$, and $s_1 \geq s_2 \geq \dots \geq s_r > 0$.

Problem 1 — Loss Landscape Geometry (15 min)

This problem explores the critical point structure of deep linear networks, following Achour et al. (2024).

(a) Consider the **diagonal** case: $d_0 = d_1 = \dots = d_L = d$, and the teacher is diagonal, $M = \text{diag}(s_1, \dots, s_d)$, with $s_1 > s_2 > \dots > s_d > 0$. Restrict

attention to diagonal weight matrices $W_l = \text{diag}(w_l^{(1)}, \dots, w_l^{(d)})$. Show that the loss decomposes into d independent scalar problems:

$$\mathcal{L} = \frac{1}{2} \sum_{\alpha=1}^d \left(s_{\alpha} - \prod_{l=1}^L w_l^{(\alpha)} \right)^2$$

(b) For a single scalar mode with target $s > 0$, find all first-order critical points of $\ell(w_1, w_2) = \frac{1}{2}(s - w_1 w_2)^2$ at depth $L = 2$. Show that the critical points are:

- The global minimum manifold: $w_1 w_2 = s$
- The origin: $w_1 = w_2 = 0$

Classify the origin as a saddle point by computing the Hessian H of ℓ at $(0, 0)$ and showing it has both positive and negative eigenvalues.

Hint: Write the gradient equations

(c) Achour et al. show that every first-order critical point $\theta = (W_1, \dots, W_L)$ satisfies: there exists a subset $S \subseteq \{1, \dots, d\}$ such that

$$W = W_L \cdots W_1 = P_S M$$

where $P_S = U_S U_S^\top$ is the orthogonal projector onto the span of the left singular vectors $\{u_{\alpha}\}_{\alpha \in S}$ of M . For the diagonal case $M = \text{diag}(s_1, \dots, s_d)$, write the value of the loss at the critical point corresponding to $S = \{1, \dots, r\}$ with $r < d$.

(d) The group $GL_h := GL_{d_1} \times \cdots \times GL_{d_{L-1}}$ acts on the weights by:

$$(W_1, \dots, W_L) \mapsto (g_1 W_1, g_2 W_2 g_1^{-1}, \dots, W_L g_{L-1}^{-1})$$

Verify that the student map $\mu(\theta) = W_L \cdots W_1$ is invariant under this action: $\mu(g \cdot \theta) = \mu(\theta)$.

Problem 2 — Gradient Flow and Conserved Quantities (20 min)

We now study gradient flow: $\dot{\theta}(t) = -\nabla_{\theta} \mathcal{L}(\theta(t))$, the continuous-time limit of gradient descent with infinitesimal learning rate.

(a) For a two-layer diagonal DLN ($L = 2$) with loss $\mathcal{L} = \frac{1}{2} \|M - W_2 W_1\|_F^2$, assume that each weight matrix W_i is a diagonal matrix and whitened inputs. Derive the gradient flow equations for each layer. Show that:

$$\dot{W}_1 = W_2^\top (M - W_2 W_1), \quad \dot{W}_2 = (M - W_2 W_1) W_1^\top$$

(b) Define the **balancedness matrix**:

$$G := W_2^\top W_2 - W_1 W_1^\top$$

Show that G is conserved under the gradient flow, i.e. $\dot{G} = 0$. In other words, gradient flow is constrained on the balanced manifold

$$\mathcal{M}_\beta := \{\theta \in \Omega \mid W_{\ell+1}^\top W_{\ell+1} - W_\ell W_\ell^\top = \beta_\ell, \ell = 1, \dots, L-1\}.$$

Hint: Compute $\dot{G}$, substitute the gradient flow equations and verify that the terms cancel pairwise.

(c) From now on, assume **balanced initialization**: $G(0) = 0$, which by part (b) means $W_2^\top W_2 = W_1 W_1^\top$ for all time. We want to derive the gradient flow in function space (the ODE for the student $W = W_2 W_1$). This requires several steps.

(c.i) Compute $\dot{W} := \frac{d}{dt}(W_2 W_1)$ using the gradient flow equations from (a). Show that:

$$\dot{W} = (M - W) W_1^\top W_1 + W_2 W_2^\top (M - W)$$

(c.ii) We now need to express $W_1^\top W_1$ and $W_2 W_2^\top$ in terms of $W = W_2 W_1$. For simplicity, assume that the weight matrices W_l are diagonal. Show that:

$$W_1^\top W_1 = (W^\top W)^{1/2}$$

(c.iii) Similarly, show that $W_2 W_2^\top = (W W^\top)^{1/2}$.

(c.iv) Substitute the results of (c.ii) and (c.iii) into (c.i) to obtain:

$$\dot{W} = (W W^\top)^{1/2} (M - W) + (M - W) (W^\top W)^{1/2}$$

(d) Define the NTK operator for $L = 2$ as:

$$K[F] := (W W^\top)^{1/2} F + F (W^\top W)^{1/2}$$

Show that the gradient flow from (c) can be written as $\dot{W} = K[M - W]$.

It turns out that this result generalizes to depth L on the balanced manifold and to non-diagonal weight matrices:

$$\dot{W} = \sum_{k=1}^L (W W^\top)^{\frac{L-k}{L}} (M - W) (W^\top W)^{\frac{k-1}{L}}$$

This is the NTK equation in the case of DLNs. The NTK equation is a gradient flow in function space with NTK being a preconditioning operator for the gradient.

Problem 3 — Rich Regime: saddle-to-saddle (25 min)

This is the core problem. We derive the exact solution of the rich regime dynamics directly from the self-consistent equation of Problem 2, emphasizing the NTK perspective.

(a) Alignment assumption. Start from the balanced gradient flow equation derived in Problem 2(c):

$$\dot{W} = (WW^\top)^{1/2}(M - W) + (M - W)(W^\top W)^{1/2}$$

We work in the **rich regime** (small initialization) with balanced weights. Assume that the student is **aligned** to the teacher: the singular vectors of $W(t)$ coincide with those of M at all times. Concretely, write:

$$M = U \operatorname{diag}(s_1, \dots, s_d) V^\top, \quad W(t) = U \operatorname{diag}(w_1(t), \dots, w_d(t)) V^\top$$

where U, V are the left/right singular vectors of M , and $w_\alpha(t) \geq 0$ are the evolving singular values of W .

(a.i) Under this alignment assumption, show that the NTK operator is aligned to the task, i.e. $K[F]$ preserves the SVD basis. In particular, show that:

$$(WW^\top)^{1/2} = U \operatorname{diag}(w_1, \dots, w_d) U^\top, \quad (W^\top W)^{1/2} = V \operatorname{diag}(w_1, \dots, w_d) V^\top$$

and that the residual is $M - W = U \operatorname{diag}(s_1 - w_1, \dots, s_d - w_d) V^\top$.

Hint: Recall that for $A = U \operatorname{diag}(\sigma_i) V^\top$, we have $AA^\top = U \operatorname{diag}(\sigma_i^2) U^\top$, and therefore $(AA^\top)^{1/2} = U \operatorname{diag}(|\sigma_i|) U^\top$.

(a.ii) Substitute into the self-consistent equation. Show that the matrix equation decouples into d independent scalar ODEs:

$$\dot{w}_\alpha = 2 w_\alpha (s_\alpha - w_\alpha), \quad \alpha = 1, \dots, d$$

Hint: Compute the product $(WW^\top)^{1/2}(M - W)$ in the SVD basis. It is diagonal with entries $w_\alpha(s_\alpha - w_\alpha)$. The second term contributes identically, giving the factor of 2.

(b) Timescale of learning. The ODE $\dot{w} = 2w(s - w)$ is a logistic equation whose solution is:

$$w(t) = \frac{s}{1 + \left(\frac{s}{w_0} - 1\right) e^{-2st}}$$

where $w_0 = w(0)$. Compute the time t_α it takes for mode α to travel from initial strength w_0 to a final strength w_f . Show that:

$$t_\alpha = \frac{1}{2s_\alpha} \ln \left(\frac{w_f (s_\alpha - w_0)}{w_0 (s_\alpha - w_f)} \right)$$

Deduce that modes with larger singular values s_α are learned faster. This is a **separation of timescales**: the network learns features in decreasing order of their singular value strength.

(c) Incremental learning. Consider a teacher with r nonzero singular values and a small uniform initialization $w_\alpha(0) = w_0 \ll s_r$ for all α . Qualitatively, discuss:

- What is the approximate distribution of singular values of $W(t)$ at early times?
- How do the singular values of $W(t)$ evolve throughout training?
- What happens in the limit $t \rightarrow \infty$?

What does this analysis tell us about the sequence of critical points visited by the gradient flow?

Problem 4 — Lazy Regime (15 min)

We now show that large initialization freezes the NTK and eliminates the timescale separation found in Problem 3. For clarity, we work with the **diagonal scalar** model from Problem 1(a), so each mode α is independent. This avoids matrix algebra and isolates the essential mechanism.

Setup. Consider a single mode: a depth-2 diagonal DLN with scalar weights a, b learning a target $s > 0$. From Problems 2–3, the balanced gradient flow for $w = ab$ is:

$$\dot{w} = 2w(s - w)$$

The factor $2w$ is the (scalar) NTK — it is **state-dependent**: the effective learning rate depends on the current value of w .

Now initialize at a **large** value $w_0 \gg s > 0$. Assume that in the early phase of training, while w has not changed much from w_0 , the ODE becomes approximately:

$$\dot{w} \approx 2w_0(s - w)$$

(b) Frozen-NTK solution. Solve the linearized ODE from (a). Show that:

$$w(t) = s + (w_0 - s)e^{-2w_0 t}$$

What is the learning timescale?

(c) No timescale separation. Now restore the mode index α . In the lazy regime, each mode satisfies $\dot{w}_\alpha \approx 2w_0(s_\alpha - w_\alpha)$ with the **same** rate $2w_0$ for all α . Compare the lazy learning time t_α^{lazy} with the rich-regime timescale from Problem 3(b): $t_\alpha^{\text{rich}} \propto 1/s_\alpha$.

Problem 5 — Mixed Dynamics: Unifying Lazy and Rich (15 min)

In practice, networks are neither purely lazy nor purely rich. Tu, Aranguri & Jacot (2024) provide a unified description. We develop the key ideas using the diagonal scalar model.

(a) The interpolating ODE. In Problems 3 and 4, we studied the same ODE $\dot{w}_\alpha = 2w_\alpha(s_\alpha - w_\alpha)$ in two limits: small w_0 (rich) and large w_0 (lazy). A more general model for a depth-2 DLN with balanced initialization at scale σ and width n replaces the scalar NTK $2w_\alpha$ by:

$$\dot{w}_\alpha = 2\sqrt{w_\alpha^2 + \tau^2} (s_\alpha - w_\alpha)$$

where $\tau := \sigma^2\sqrt{n}$ is a **threshold** parameter that depends on the initialization scale and width.

Verify that this ODE reproduces the two known regimes:

- **Rich limit** ($\tau \rightarrow 0$): recover $\dot{w}_\alpha = 2|w_\alpha|(s_\alpha - w_\alpha)$.
- **Lazy limit** ($\tau \rightarrow \infty$ with w_α bounded): recover $\dot{w}_\alpha \approx 2\tau(s_\alpha - w_\alpha)$.

(b) Two-phase dynamics. At initialization, all modes start at $w_\alpha(0) \sim \sigma^2 \ll \tau$ (since $\sqrt{n} \gg 1$). Observe that:

- When $|w_\alpha| \ll \tau$, the effective learning rate is $\approx 2\tau$, the same for all modes (lazy behavior).
- When $|w_\alpha| \gg \tau$, the effective learning rate is $\approx 2|w_\alpha|$, which is mode-dependent (rich behavior).

Describe the resulting two-phase dynamics in words: what happens first, and what happens later?

(c) Mode-by-mode transition. Not all modes cross the threshold τ at the same time: which modes cross it first? Explain why this means that a single network can simultaneously have some modes in the rich regime and others still in the lazy regime.

Problem 6 (Bonus) — Stochastic Implicit Bias (10 min)

SGD introduces noise from mini-batching. Write the SGD update as a combination of gradient and noise term (use a LLM to check your answer). A continuous model is the Langevin SDE:

$$d\theta_t = -\nabla \mathcal{L}(\theta_t) dt + \sqrt{\eta \Sigma(\theta_t)} dB_t$$

where $\Sigma(\theta)$ is the covariance of the stochastic gradient noise and η is the learning rate.

(a) The probability density $p(\theta, t)$ of the parameters evolves according to the Fokker–Planck equation:

$$\partial_t p = -\nabla \cdot \mathbf{j}, \quad \mathbf{j} = -\nabla \mathcal{L}(\theta) p(\theta) - \frac{\eta}{2} \nabla \cdot (\Sigma(\theta) p(\theta))$$

where $\mathbf{j}$ is the probability current. Assume: (i) stationarity $\partial_t p^* = 0$, (ii) thermal equilibrium $\mathbf{j} = 0$, and (iii) isotropic noise $\Sigma = \sigma^2 I$. Show that the equilibrium distribution is the Boltzmann distribution:

$$p^*(\theta) \propto \exp\left(-\frac{2}{\eta\sigma^2} \mathcal{L}(\theta)\right)$$

Hint: Setting $\mathbf{j} = 0$ with $\Sigma = \sigma^2 I$ gives $\nabla \mathcal{L} p + \frac{\eta\sigma^2}{2} \nabla p = 0$. This is a first-order ODE for p in terms of $\mathcal{L}$. Try the ansatz $p \propto e^{-\beta \mathcal{L}}$ and solve for β .

(b) The ratio $\frac{2}{\eta\sigma^2}$ plays the role of an inverse temperature β . Interpret what happens to the equilibrium distribution when:

- η is very small (low temperature)
- η is very large (high temperature)

Which regime favors flatter minima, and why might this be beneficial for generalization?

(c) In practice, SGD noise is **not** isotropic: $\Sigma(\theta)$ depends on both the loss landscape and the data. Without solving anything, explain qualitatively why anisotropic noise can introduce an implicit bias that goes beyond what the loss

function $\mathcal{L}$ alone would select. Specifically, why might SGD preferentially escape sharp directions of the loss while remaining stable along flat directions?